

GENERAL SPECIFICATION G3.12

## Soil Stabilisation Works

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## **G3.12 SOIL STABILISATION WORKS**

### **G3.12.1 General**

- (a) The Contractor's responsibilities for any damage to persons or properties shall be in accordance with the Conditions of Contract and the Section "Preliminaries and General".
- (b) This Specification is for application of soil stabilisation to achieve the following:
  - (i) Formation of a layer to resist base heave within excavations for shafts and trenches;
  - (ii) Strengthen the ground and prevent soil heave at manhole/ chamber/ open cut trench locations;
  - (iii) Protection against water flow and soil loss at all manholes, chambers, jacking/receiving shafts and open cut trenches; and
  - (iv) Prevent surrounding ground movement caused by construction activities.
- (c) This section specifies certain materials and workmanship requirements for the construction of jet grout piles as the means of stabilising the sub-soil for the applications mentioned above. The Contractor is to note that jet grouting is meant for the applications mentioned above and is not specified as a replacement to foundation works and timbering. The Contractor is fully responsible for the effectiveness of the jet grouting works and any approval to carry out the works given by the S.O. will not absolve the Contractor of his responsibility towards the work.
- (d) The costs for all ground treatment and/or soil stabilisation works for working shafts needed to construct the sewers and manholes, to be carried out by the Contractor and as designed by the Contractor's Professional Engineer (Geotechnical, if applicable), shall deem to be included in the Contractor's pricing for the construction of the sewers and manholes.
- (e) All Professional Engineers must have a current valid practicing certificate.

### **G3.12.2 Design and Supervision**

- (a) The Contractor shall engage a Professional Engineer (Geotechnical, if applicable) to carry out a detailed study including any additional soil investigation deemed necessary of the whole Site as specified to identify sites that require soil stabilisation. The Professional Engineer (Geotechnical, if applicable) engaged by the Contractor shall be responsible for the design, supervision and submission of reports for all soil stabilisation works as specified, during the whole duration of the Contract. The Contractor shall submit a detailed report endorsed by the Professional Engineer (Geotechnical, if applicable) of the study with recommendations on all sites that require soil stabilisation within two (2) months of Award of Tender. An item has been allowed for in the Bills of Quantities for the engagement of Qualified Personnel for the whole duration of the Contract, to carry out the works as mentioned above. The rate shall also include for all the necessary additional soil investigation works.

- (b) The Contractor is to note that the S.O. may direct soil stabilisation works at certain sites and a provisional item has been allowed in the Bills of Quantities for the Contractor to price for the soil stabilisation work. Notwithstanding this, the Contractor shall be responsible for protecting all surrounding properties and services throughout his work. The rate shall also be deemed to include for all additional measures that the Contractor shall undertake to ensure that the jet grouting works so carried out by him is functional for the applications mentioned above.
- (c) The Contractor's Professional Engineer (Geotechnical, if applicable) is to carry out the detailed design of all soil stabilisation works including those sites directed by the S.O. (where applicable), submit method statements, submit trial test and final reports and to supervise the whole of the soil stabilisation works. All costs relating to this is deemed to be included in the rates for the engagement of the Professional Engineer (Geotechnical, if applicable) as mentioned above.

### **G3.12.3 Soil Stabilisation using Jet Grout Piles**

- (a) Jet Grout Pile Construction
  - (i) Jet grouting is a technique whereby a mix of cement, water, air and, optionally, chemical additives is injected continuously at high pressure from a nozzle which is progressively removed from below the ground from a depth to which it had previously been bored. The grout, which has been injected under pressure, mixes with and replaces the sub-soil. When it hardens, the grout forms a column of hardened material. A series of columns are formed at a close spacing to form a hardened layer;
  - (ii) The recommendations of SS EN 1997: "Eurocode 7: Geotechnical Design" and BS EN 12716: "Execution of Special Geotechnical Works. Jet grouting" shall apply to these works in so far as they are applicable to jet grout pile construction, except that, in the event of conflict or ambiguity, this Specification shall take precedence and as directed by the S.O.. The Contractor shall allow for any difficulty and inconvenience or site constraints to be encountered during construction and is held to have visited the Site to ascertain the existing condition. The Contractor may propose alternative method of construction if the conditions of the Site is deemed not suitable for pipe-jacking method. The Contractor shall consult and obtain the approval from the S.O. and all relevant authorities before commencement of works.
- (b) Method Statement
  - (i) The Contractor shall submit a comprehensive method statement, endorsed by the Contractor's Professional Engineer (Geotechnical, if applicable), giving full details of the materials, plant and operations involved in the different types of jet grout pile applications. The Method Statement shall include the following details: -
    - 1) Location of the jet grout piles;
    - 2) Dimensions of the jet grouted layer including founding level, layer thickness and cut-off level;
    - 3) Dimensions of the grout pile columns including theoretical diameter and spacing;

- 4) The sequence of forming the jet grout piles and the sequence of piling;
- 5) The method of monitoring and checking the stability of neighbouring properties, roads, services and other underground structures;
- 6) The method of monitoring the location, depth and verticality of the jet grout nozzle, including a plan showing the grid of the jet grout pile centres;
- 7) The Specifications and details of the following equipment to be used:
  - Grout mixers;
  - Injection pumps;
  - Injection nozzles; and
  - Nozzle drilling equipment.
- 8) The method of measuring and controlling the following parameters:
  - Injection pump pressure;
  - Injection nozzle rod rotational speed;
  - Injection air pressure; and
  - Nozzle level.
- 9) The method of drilling and contingencies for different conditions which may be encountered.
- 10) The method of mixing and pumping the grout including:
  - Cement type, properties and test certificates;
  - Mix proportions;
  - Technical details and test reports on chemical admixtures (if any);
  - Trial mix details;
  - Details of the storage and mixture of the elements; and
  - Testing and monitoring methods.
- 11) The method of measurement and control of surface heave and settlement due to the treatment including:
  - Trial tests;
  - Instrumentation and settlement monitoring locations and details;
  - Identification of critical buildings and sub-surface utilities; and
  - Measures to be taken to limit the effects of the treatment.
- 12) The method of containing excess grout and soil at the surface and disposal of the spoil;
- 13) The methods of verification and measurement of the grout strength and zone of treatment; and

- 14) A comprehensive risk analysis which shall identify any safety hazards arising from the works and describe measures that will be taken to mitigate the risks.
- (ii) Construction of the jet grout piling works shall not commence until the Contractor's proposals have been accepted by the S.O.. The Contractor shall then carry out the jet grouting work, under the supervision of the Professional Engineer (Geotechnical, if applicable) engaged by him, strictly in accordance with the design parameters adopted by the Professional Engineer (Geotechnical, if applicable).
- (c) Levels of Work
- (i) Jet grout piles shall be constructed to the base levels of all manholes/ chambers/ open cut trenches as specified to achieve the objectives as set out under Clause "General" of this Specification.
- (d) Zone of Treatment
- (i) The plan area of the zone to be jet grouted shall be defined in the Contractor's Professional Engineer (Geotechnical, if applicable) design and method statement and shall be subject to the acceptance of the S.O.. The zone may be reviewed by the S.O. depending on the conditions encountered during the treatment works. Grouting shall be carried out such that the entire specified zone is treated.
- (e) Tolerances
- (i) Construction shall be carried out in accordance with the following tolerances:
- 1) The jet grout piles shall be vertical to within a tolerance of 1:50; and
  - 2) The jet grout piles shall be located with a plan tolerance of plus or minus 100mm of the location shown on the plan of the pile grid.
- (ii) If, during the testing, the above tolerances are shown to have been exceeded, the Contractor shall propose and carry out the following at his own costs:
- 1) Further testing of the jet grout layer to verify that there are no gaps in the treatment; and
  - 2) If there are any gaps, submit proposals endorsed by the Contractor's Professional Engineer for rectifying the defects.
- (iii) If the tolerances are exceeded consistently, the S.O. may call upon the Contractor to submit for his approval an amended method statement for all future jet grout piling works. All works ordered or done under this sub-clause shall be at the Contractor's own expense and no extension of time will be granted for such works.
- (f) Headroom
- (i) The Contractor shall note any restricted headroom on portions of the Works and shall satisfy the S.O. that his method of construction of the jet grout piles can be accommodated within these restrictions.

(g) Grout

- (i) The grout to be used shall consist of a mixture of cement, water and, if necessary, chemical admixtures. The proportion of each constituent shall be specified in the Contractor's method statement together with a target strength for the grout mix and a target strength for the completed treated layer;
- (ii) Cement shall be Ordinary Portland cement complying with SS EN 197: "Cement" but containing not less than 4% and not more than 13% proportion by weight of tricalcium aluminate;
- (iii) Suitable admixtures may be used in the grout mix with the prior approval of the S.O.. The type and source of the admixture, amount to be added and method of use shall be to the acceptance of the S.O. who shall be provided with the following data:
  - 1) The manufacturer's recommended dosage and detrimental effects of under dosing and over dosing;
  - 2) The chemical name(s) of the main active ingredient(s) in the admixture;
  - 3) Whether or not the admixture contains chlorides and, if so the chloride content of the admixture expressed as a percentage of equivalent anhydrous calcium chloride by weight of admixture;
  - 4) Whether or not the admixture leads to the entrapment of air when used at the manufacturer's recommended dosage;
  - 5) Information on the toxicity of the admixture; and
  - 6) Evidence of previous satisfactory usage of the admixture in grout for jet grouting works.
- (iv) The grout and admixtures used must be chemically inert, non-pollutive and present no environmental risks;
- (v) Prior to the commencement of the work, test cubes for the grout mixture shall be made and tested in accordance with BS 1881: "Testing Concrete" and BS EN 12390: "Testing hardened concrete". Eight test cubes shall be taken and half are to be tested at 7 days with the remainder tested at 28 days to verify the target compressive strength of the grout mix.

(h) Jet Grouting – Trial Test

- (i) Well in advance of the planned commencement date for the jet grouting works the Contractor shall conduct a trial of the jet grouting method to prove the suitability of the method. An item has been allowed for in the Bills of Quantities for the Contractor to price for the trial of the jet grouting;
- (ii) The trial shall be conducted using the materials and equipment to be used in the actual jet grouting works;
- (iii) The trial shall demonstrate that the procedures used will produce a homogeneous grouted pile of the required diameter;
- (iv) The location and the depth of the jet grouting for the trial shall be proposed by the Contractor for acceptance by the S.O.;

- (v) Prior to the commencement of the trial the Contractor shall prepare for the acceptance of the S.O. a method statement specifically for the trial. The trial shall not commence until the S.O. accepts the method statement. The method statement shall set out clearly the following:
- 1) The objective of the trial;
  - 2) Any deviations from the method statement for the jet grouting works and the reasons for the deviations;
  - 3) Tests to be performed and the testing procedure. The tests shall include methods of verification of the homogeneity of the grouted piles and shall include:
    - Visual inspection of at least two piles which shall be constructed to about 1.5m of the ground surface and exposed by careful excavation; and
    - Coring of at least one pile at a location on the pile to be determined by the S.O., together with standard penetration tests at intervals of not greater than one metre. At least two samples for each core shall be tested by unconfined compression tests.
  - 4) After completion of the tests, the Contractor shall submit to the S.O. a full report containing the results of all tests carried out during the trial. The report shall conclude with the column diameter and spacing, grout mix design, grouting pressure, relief hole spacing etc., to achieve the jet grouting layer specified in the Contractor's design. The jet grouting works proper shall not commence until the S.O. has approved the trial test report; and
  - 5) If the trial test fails, the Contractor's Professional Engineer (Geotechnical, if applicable) shall re-evaluate the design of the jet grout piles and the Contractor is to carry out another trial test. All works ordered or done under this sub-clause shall be at the Contractor's own expense and no extension of time will be granted for such works.
- (i) Tests and Testing
- (i) The Contractor's Professional Engineer (Geotechnical, if applicable) shall propose in the method statement the type and frequency of tests to be carried out to verify the uniformity of the final jet grouting layer and its compliance with the design criteria. All costs for carrying out the tests shall be deemed to be included in the pricing for jet grouting works;
  - (ii) The tests shall consist of the following:
    - 1) Coring of the completed jet grouting mass at a frequency of no less than one core for every 100m<sup>2</sup> of plan area of treated soil or 1% of the number of treated columns, whichever is the greater. The coring shall be of 50mm diameter;
    - 2) Compressive strength tests on the core samples; and
    - 3) Standard penetration tests or cone penetration tests at a frequency to be proposed by the Contractor for the acceptance of the S.O..

- (iii) If the testing shows any defects or deficiencies in the jet grouting treatment the Contractor shall propose for the S.O.'s acceptance measures that will be taken to correct the deficiencies. All works ordered or done under this sub-clause shall be at the Contractor's own expense and no extension of time will be granted for such works.
- (j) Records
  - (i) The Contractor shall submit to the S.O. daily records of the previous day's work. The daily records shall include the following information:
    - 1) The equipment being used including the diameter of the drilling rod and the nozzle sizes;
    - 2) The jet grouting column nominal diameter, spacing, mix, pressure and other relevant equipment settings;
    - 3) The location and reference number of each of the columns grouted;
    - 4) The levels of the beginning and ending of the grouting for each column;
    - 5) The grout mix proportions and any changes in the materials used;
    - 6) Details of any tests taken;
    - 7) Monitoring results as specified in the Clause "Ground Movements" of this Specification;
    - 8) A log of the soil type encountered so far as can be ascertained from the drilling and the spoil; and
    - 9) The volume of grout used.
  - (ii) Upon completion of the jet grouting works in each area, the Contractor shall submit to the S.O. two (2) copies of the final report, endorsed by the Contractor's Professional Engineer (Geotechnical, if applicable), on the results of the treatment which shall include but not limited to the following:
    - 1) A full set of all test results; and
    - 2) A description of any deviations from the design giving reasons for the deviations and proposals for any remedial works.
- (k) Soil Stabilisation Using Fissure Grouting
  - (i) Fissure grouting is defined as the filling of naturally occurring grouting voids or cavities adjacent to the works, whether underground water pressure or not. It does not include artificially induced fissures such as compensation grouting. Grout injection shall be by cement grout or other materials;
  - (ii) The recommendations of SS EN 1997: "Eurocode 7: Geotechnical Design" and BS EN 12715: "Execution of Special Geotechnical Work. Grouting" shall apply to these works in so far as they are applicable to soil stabilisation using fissure grouting, except that, in the event of conflict or ambiguity, this Specification shall take precedence and as directed by the S.O.;

- (iii) Holes shall be drilled to 45 mm nominal diameter, unless otherwise approved, at locations and to the angles of inclination and depths as approved by the S.O.. Where agreed with the S.O., holes shall be water tested in accordance with the Specification. The grouting pressure shall be determined by the Contractor taking account of tests, if any, and the nature of the ground, overburden depth, nature and location of other parts of the Works and structures and property of others;
  - (iv) All grout holes shall be flushed clean with clean water under continuous pressure up to the grouting pressure;
  - (v) Cement grout injection shall commence with a mix of specified water: cement ratio and shall continue until refusal, defined as 20 litres or less in a ten-minute period at the agreed pressure, or until a maximum amount of grout specified for that mix has been injected. Where refusal has not been achieved, holes shall be re-drilled or further holes drilled and grout injection restarted;
  - (vi) If during the grouting of any hole, grout is found to flow from points in the rock surface, or from the hole, such leaks shall be plugged, caulked or stemmed;
  - (vii) Where a hole is required to be re-drilled, this shall be done as soon as possible after initial set of grouts in that hole. Bentonite or other additives may be added to the final grout mix to facilitate re-grouting subject to the agreement of the S.O.;
  - (viii) Upon completion of grouting operations, the Contractor shall drill test holes at positions, orientation and depths agreed with the S.O.. Such test holes may be required to be water tested. Consequent upon the results of tests, the grouting may be accepted or further drilling or grouting required; and
  - (ix) Normal tunnelling operations shall not commence until the injected grout has set sufficiently.
- (I) Soil Stabilisation Using Compensation Grouting
- (i) Compensation grouting consists of the introduction into the ground of grout layers to compensate for normal tunnelling settlement and to control ground and existing structure movements;
  - (ii) The recommendations of SS EN 1997: “Eurocode 7: Geotechnical Design” and BS EN 12715: “Execution of Special Geotechnical Work. Grouting” shall apply to these works in so far as they are applicable to soil stabilisation using compensation grouting, except that, in the event of conflict or ambiguity, this Specification shall take precedence and as directed by the S.O.;
  - (iii) Grouting shall be executed in accordance with of this Specification;
  - (iv) Monitoring of compensation grouting shall be in accordance with the Section “Specifications for Instrument and Monitoring Works”; and
  - (v) Continual assessment and reappraisal of the effects on the ground and structures of compensation grouting shall be carried out by the Contractor and agreed with the S.O.’s Amendments shall be made to the grouting procedures to maintain the designed control.

(m) Soil Stabilisation Using Deep Soil Mixing (DSM)

(i) Deep Soil Mixing Works

- 1) Deep Soil Mixing (DSM) is a process of improving the ground by cutting a column of soils and mixing it mechanically using a rotating mixing tool with lime or cement. The DSM column can be formed by either wet mixing method or dry mixing method; and
- 2) The recommendations of SS EN 1997: "Eurocode 7: Geotechnical Design" and BS EN 14679: "Execution of Special Geotechnical Works. Deep Mixing" shall apply to these works in so far as they are applicable to soil stabilisation using deep soil mixing, except that, in the event of conflict or ambiguity, this Specification shall take precedence and as directed by the S.O..

(ii) DSM – Trial Tests

- 1) Well in advance of the planned commencement date for the DSM works the Contractor shall conduct a trial of the DSM to prove the suitability of the method. An item has been allowed for in the Bills of Quantities for the Contractor to price for the trial. The Contractor shall prepare for the acceptance of the S.O. a method statement specifically for the trial. The trial shall not commence until the S.O. accepts the method statement. The method statement shall set out clearly the following:
  - The objective of the trial;
  - Any deviations from the method statement for the jet grouting works and the reasons for the deviations; and
  - Tests to be performed and the testing procedure.
- 2) The trial shall consist of a minimum of four columns formed at the depth and in similar soil conditions as the proposed DSM. The trial shall be used to confirm the following: -
  - Diameter of the DSM columns formed;
  - Properties, especially the compressive strength and stiffness, of the DSM columns;
  - Applicability of the operational parameters;
  - Type and quality of binder to achieve the performance requirements; and
  - Effect of the DSM columns on the soils above the treatment zone.
- 3) The effectiveness of the trial shall be demonstrated with a minimum of two boreholes used to obtain cores through full depth of the DSM block. The boreholes shall be targeted with the midway between the centre and outer surface of the columns. The core shall be fully logged and shall be tested for strength and stiffness.

- 4) The Contractor shall take measures to avoid the reduction of strength and stiffness of the soils between the ground level and the top of the treated zone. The boreholes used to test the treated zone shall be used to measure the strength and stiffness of the soils between the ground level and the top of the treated zone to demonstrate the effectiveness of the measures.
- 5) After completion of the tests, the Contractor shall submit to the S.O. a full report containing the results of all tests carried out during the trial. The DSM works proper shall not commence until the S.O. has approved the trial test report.
- 6) In the event where the trial test fails, the Contractor's Professional Engineer (Geotechnical, if applicable) shall re-evaluate the design of the DSM columns and the Contractor is to carry out another trial test. All works ordered or done under this sub-clause shall be at the Contractor's own expense and no extension of time will be granted for such works.

(iii) Installation

- Before installation of the columns, the columns position shall be marked;
- The mixing materials used shall be non-toxic. The material data sheets for each material shall be submitted to the S.O. for approval; and
- All drilling and grouting effluent shall be collected in trenches, pits or tanks and they are not permitted to spread over or outside the site. Measures shall be taken to ensure that the effluent does not enter the public system. All effluent shall be removed from the Site and disposed of in compliance with the regulations.

(iv) Records

- 1) During the DSM column installation, the Contractor shall submit daily progress reports and record sheets on the DSM works. The record on data sheets shall include the following:
  - Type of equipment and material used;
  - Quantity of cement or lime used per meter of DSM column;
  - Total quantity of cement and lime used in the day;
  - Length of DSM columns installed;
  - Duration and time of works; and
  - Operational parameters.
- 2) Upon completion of the DSM works in each area, the Contractor shall submit to the S.O. two (2) copies of the final report, endorsed by the Contractor's Professional Engineer (Geotechnical, if applicable). The report shall include, but not limited, to the following:
  - A full set of all test results; and
  - A description of any deviations from the design giving reasons for the deviations and proposals for any remedial works.

- (v) Soil Stabilisation Using Chemical Grouting or 'Water Glass' ground treatment
- 1) The chemical grouting of 'water glass' ground treatment is commonly used to reduce the permeability of the sand layers and hence stabilize the zone directly in contact with the soft eye of the launching shaft.
  - 2) Process (for the Contractor to comply)
    - The mixed chemical solutions A & B will be transferred to the double tube injection pipe located at the base of the shaft by high pressure hose. The hose is housed in a PVC pipe to contain any leaks.
    - The chemicals used shall be Sodium Silicate and Sulphuric Acid. The Sodium Silicate is transported in 1.0m<sup>3</sup> IBC (Industrial Bulk Container) supported by a rigid Aluminium frame. The Contractor shall take due care to store it in a protected enclosure and on to the sand lined steel tray.
    - The Sulphuric Acid is transported in plastic 20 litres containers for easy handling. The containers are carried to the temporary storage area and placed on the sand lined tray. A maximum of 5 no. 20 litres containers are stored temporarily on site during working hours. The Contractor shall note that at the end of each shift all containers (whether empty or full) are to be removed from site.
    - The Contractor shall prepare the 'B' solution first by mixing water and sodium silicate.
    - The acid is transferred from the 20 litre container by an electric suction pump into the plastic measuring container. Once the predetermined amount of acid has been transferred, the acid is then released into the mixing container that contains water and sodium silicate ('B' solution). The mixture of acid and 'B' solution combine to form the 'A' solution.
  - 3) Checking Drilling
    - After the ground treatment works has been carried out, the Contractor shall carry out a check to assess the permeability of the sand layer by means of drilling. The check drilling location to be submitted the S.O. for approval. The permeability values to be reduced to minimum  $2.0 \times 10^{-7}$  m/sec.
  - 4) Records
    - During the Chemical grouting column installation, the Contractor shall submit daily progress reports and record sheets on the works. The record on data sheets shall include the following:
      - Type of equipment and material used;
      - Flow rate and permeability calculation;
      - Grouting /injection data;
      - Summary grouting/injection data;
      - Diagram of chemical grouting; and

- Operational parameters.
- Upon completion of the works in each area, the Contractor shall submit to the S.O. two (2) copies of the final report, endorsed by the Contractor's Professional Engineer. The report shall include, but not limited, to the following:
  - A full set of all test results; and
  - A description of any deviations from the design giving reasons for the deviations and proposals for any remedial works.
- Contractor's responsibilities for any damage to persons or property in relation to the soil stabilisation works shall be in accordance with the Conditions of Contract.

#### **G3.12.4 Ground Movements**

- (a) The Contractor's responsibilities for any damage to persons or property in relation to the soil stabilisation works shall be in accordance with the Conditions of Contract.
- (b) Prior to starting work, all sub-surface utilities shall be located and protected to the satisfaction of the appropriate authority.
- (c) Soil stabilisation shall be carried out with the utmost care to prevent disturbances to the nearby services and properties. Care shall be taken to prevent undesirable ground movements. Care shall also be taken to prevent intrusion of grout or other material into services or properties as a result of the soil stabilisation works. The Contractor shall be responsible to rectify any damage to services as a result of intrusion of grout or other material to the satisfaction of the S.O. or other relevant authorities. No claims whatsoever shall be entertained for the lack of such knowledge or otherwise.
- (d) The Contractor shall establish suitable temporary bench marks for the purposes of monitoring ground movements during ground treatment. Such bench marks shall be outside the zone of influence of the soil stabilisation works.
- (e) The Contractor's attention is brought to the Section "Specifications for Instrument and Monitoring Works". Over and above the requirements set out in the above-mentioned Section, the Contractor shall monitor ground movements due to the soil stabilisation works at least twice a day. All readings and measurements shall be properly taken, recorded and interpreted by qualified personnel, endorsed by the Professional Engineer and one set of reports shall be submitted to the S.O. daily.
- (f) In the event of undesirable ground movements, the Contractor shall immediately stop treatment works and propose alternative treatment methods for the acceptance of the S.O.. It is the Contractor's responsibility to ensure that all surrounding properties near the Site such as the road and pedestrian bridges, the utilities installations, road facilities, foundation structures of building, underground services, conservation building, etc. are not disturbed or damaged due to soil stabilisation works. The Contractor shall be responsible to protect all the surrounding properties throughout his operations. No claims whatsoever shall be entertained for lack of such knowledge or otherwise.

**G3.12.5 Obstructions**

- (a) Upon encountering any sub-surface obstruction which prevents the soil stabilisation equipment from reaching the specified level of the treated layer, the Contractor shall immediately inform the S.O. and shall review and propose to the S.O. the steps to be taken to ensure that the ground is treated satisfactorily.

**G3.12.6 Disposal of Soil**

- (a) The Contractor shall construct bunds or walls, as necessary, to contain excess grout and soil from the soil stabilisation works. The Contractor shall also construct sediment settlement tanks to extract grout and soil sediment from any water disposed to the surface drainage system.
- (b) The spoil and grout shall be disposed of at the appropriate dumping site, approved by the S.O..

**G3.12.7 Temporary Sites Required for Soil Stabilisation Works**

- (a) The Contractor shall be responsible for obtaining approval from the relevant authorities for all temporary site areas required for carrying out the soil stabilisation works. All temporary occupation licenses, rental charges and related costs because of this shall be borne entirely by the Contractor. The Contractor shall be deemed to have allowed for this in his rates.

**END OF SECTION**